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Projet de mention

Comparison of Deep Learning methods using Knowledge Distillation to improve multi-omics integration in cancer classification

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January 29, 2026

Mention Healthcare et Services Biomédicaux

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Acknowledgements

We would like to warmly thank our supervisor on the CNRGH side, Mary Savino, for her continued availability and guidance throughout the ongoing course of this project. Her regular follow-ups have been invaluable in helping us keep track of our progress and structure our work at this stage. She has also kindly introduced us to the biological and methodological context of the project, notably by explaining the biological interpretation of the different omics modalities (such as transcriptomics and DNA methylation), as well as their appropriate use in data analysis, discussing relevant approaches for survival analysis in such datasets, and clarifying several technical aspects of the methods and articles on which this work is based.

In addition, her guidance encouraged us to develop a critical yet constructive perspective on existing approaches, helping us to question methodological choices with rigor while remaining mindful of their scope and assumptions.

We also thank the Centre National de Recherche en Génomique Humaine (CNRGH) for providing the scientific environment and resources that make this project possible, as well as for facilitating access to the data used in this study.

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Chapter 1

Introduction

This project is conducted in collaboration with Mathematics and Statistics team at the National Center for Human Genomics Research (CNRGH), a research institute based at Evry and affiliated with the CEA and Université Paris-Saclay, dedicated to large-scale human genomics. CNRGH plays a key role in the production, analysis and interpretation of high-throughput genomic and multi-omics data, with applications spanning fundamental research and biomedical studies, including oncology. The present work aligns with CNRGH's research themes on multi-omics data analysis and cancer cohorts, notably those derived from large public resources such as The Cancer Genome Atlas (TCGA)[1] and the Clinical Proteomic Tumor Analysis Consortium (CPTAC)[2] , and contributes to ongoing efforts to develop robust computational methods for exploiting complex and heterogeneous biomedical datasets.

In oncology, understanding cancer heterogeneity and disease progression requires the analysis of complex biological mechanisms occurring at multiple molecular levels. New technologies can now make it possible to generate diverse omics data, including genomics, transcriptomics, proteomics and epigenomics, each capturing complementary aspects of tumor biology. While single-omics analyses have led to important discoveries, they provide only a partial view of the underlying biological processes. This challenge is well captured by the "blind men and the elephant" metaphor (Figure 1.1), where each observer perceives only one part of a complex system—similarly, each omics modality provides a partial perspective on tumor biology, motivating integrated multi-omics modeling. Integrating multiple omics modalities has been shown to improve patient stratification, biomarker discovery

[1] https://www.cancer.gov/ccg/research/genome-sequencing/tcga

[2] https://gdc.cancer.gov/about-gdc/contributed-genomic-data-cancer-research/clinical-proteomic-tumor-analysis-consortium-cptac

and clinical outcome prediction, by jointly modeling genetic alterations, gene expression patterns, protein abundance and regulatory mechanisms. As a result, multi-omics integration has become a central methodological challenge in modern cancer research.

Figure 1.1: Blind men and the elephant" metaphor highlighting the limits of single-view observations and the motivation for multi-omics integration.[2]

In practice, multi-omics cancer cohorts are often incomplete, as not all patients have every genomic modality available, due to experimental, technical, or even economic limitations. Furthermore, omics data are characterized by a very high dimensionality compared to the small number of patients usually available in public cohorts such as TCGA. Numerous studies and methodologies have been introduced during the past decade to circumvent these issues, some of them are mentioned in the following section.

Chapter 2

Literature Review

2.1 Deep Learning for multi-omics cancer classification

2.1.1 Deep Learning for multi-omics cancer classification

Multi-omics studies combine heterogeneous modalities such as transcriptomics, methylation, miRNA, proteomics, to better capture the multifactorial mechanisms underlying cancer phenotypes and clinical outcomes. In practice, deep learning (DL) has become a key family of approaches for multi-omics cancer classification because it can learn non-linear representation from high-dimensional inputs and combine complementary signals across representations.

A common way to categorize multi-omics integration methods is based on when cross-modal information is integrated. Ballard et al. [3] distinguished three main integration strategies:

- early integration, where features from different modalities are concatenated before modeling,
- intermediate integration, where each modality is processed separately but the model learns cross-modal relationships and produces a shared latent space or integrated representation,
- late integration, where separate models are trained for each modality and their predictions are later aggregated.

This taxonomy is useful in oncology because it reflects practical trade-offs: early fusion can capture fine-grained interactions but is sensitive to missing modalities; intermediate fusion is more flexible but requires careful architectural design; late fusion is modular and often

more robust, but may fail to model cross-omic interactions explicitly. Ballard et al. [3]

further grouped recent deep learning approaches into non-generative and generative models, while highlighting the increasing use of attention mechanisms and strategies to handle incomplete data. Non-generative models learn a direct mapping from inputs to labels and are typically simpler and less computationally intensive. The authors then organized these into feed-forward networks, graph convolutional networks and autoencoder-based methods. By contrast, generative models, using variational autoencoders and Generative Adverserial Networks (GANs), aim to model aspects of the data distribution and can support representation learning and, in some settings, missing-data handling. Ballard et al. identify variational methods as a major generative family in multi-omics integration.
Beyond predictive performance, these modeling choices affect interpretability (modality

contribution or feature attribution), robustness to cohort shifts (batch effects, platform differences), and compatibility with real clinical cohorts, where complete multi-omics profiles are often unavailable.

2.1.2 Handling missing modalities in multi-omics

In multi-omics studies, incomplete or unpaired cohorts refer to settings where not every patient has all modalities measured, for instance some patient have RNA-seq and methylation data, while others have only RNA-seq. This situation is frequent in real-world cohorts due to practical constraints such as limited biological material, differences in experimental design, platform or sequencing-center variability, and cost considerations, as mentioned in the introduction.

Missingness is not a technical inconvenience, it directly affects what can be learned from the data. Flores et al. [4] emphasize the missingness mechanisms in omics datasets are often Missing At Random (MAR) or Missing Not At Random (MNAR) rather than the more convenient Missing Completely At Random (MCAR) assumption. They also note that proteomics data can exhibit substantial missingness (a large fraction of peptide values not being quantified), which can hinder downstream analyzes and model training.
A practical way to structure the literature is to distinguish three broad strategies, consistent with the perspective in Flores et al.[4] :

- **Restrict to complete cases**

The simplest approach is to keep only patients with all available modalities. However, in multi-omics this can drastically reduce the sample size and is only strictly valid under MCAR; otherwise, it can lead to biased estimates and reduced generalizability. This limitation motivates methods that aim to leverage partially observed cases instead of discarding them.

- **Imputation** (in the preprocessing or in the pipeline)
  Imputation methods fill in missing values or entire missing modality blocks. Flores et al. [4] highlight that imputation is widely used because many integration methods assume complete inputs. The main drawback is that imputation can introduce artifacts or noise, and it becomes less reliable when the number of fully observed samples is small—precisely the setting encountered in many cancer multi-omics cohorts.

- **Architectures explicitly designed for missing modalities**
  Flores et al. [4] argue that more robust approaches handle missingness directly within the modeling framework, allowing partially observed samples to contribute without requiring a separate imputation step. They describe two common design patterns: joint-imputation approaches, where missing values are inferred during model fitting (e.g., using probabilistic or generative components), and optimization-masking approaches, where the training objective is adapted to ignore missing components while still learning from observed modalities.

This third family is especially relevant for the project, as it aligns with the objective of exploiting incomplete cohorts rather than limiting analyses to complete cases.

2.1.3 Knowledge distillation in deep learning

Knowledge distillation (KD) was introduced by Hinton et al. [5] and aims to train a student model to mimic a teacher model (or an ensemble), typically by matching the teacher's soft output distribution while still learning from ground-truth labels. In this setting, class probabilities are often obtained through a temperature-scaled softmax that converts the logits z_i into a probability distribution q_i over classes:

$$q_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}. \tag{2.1}$$

Here, T denotes the temperature (with $T = 1$ recovering the standard softmax); larger values of T produce a softer probability distribution and thus provide richer similarity information between classes. Following a common formulation, the student is optimized using a combination of a supervised prediction loss and a distillation loss:

$$\mathcal{L} = \mathcal{L}_{\text{pred}}(y, z_{\text{St}}) + \gamma\, \mathcal{L}_{\text{KD}}(z_{\text{St}}, z_{\text{Te}}), \tag{2.2}$$

where z_{St} and z_{Te} are the logits produced by the student and teacher models, respectively, and γ controls the contribution of the distillation term. In practice, $\mathcal{L}_{\text{KD}}$ is often defined as the Kullback–Leibler divergence between the temperature-softened teacher and student distributions:

$$\mathcal{L}_{\text{KD}} = \text{KL}\Big(\text{softmax}\Big(\frac{z_{\text{Te}}}{T}\Big) \,\Big\|\, \text{softmax}\Big(\frac{z_{\text{St}}}{T}\Big)\Big)\,. \tag{2.3}$$

In the formulation described by Wang et al. [6], this combination of supervised and distillation objectives is used to transfer knowledge from a teacher to a student while leveraging softened outputs to improve learning in their context of medical imaging for Alzheimer's disease.

A key insight from Wang et al. [6] is that knowledge distillation can help in settings where some modalities are missing, without requiring explicit imputation. They motivate this approach by noting that many multimodal methods discard incomplete samples or impute missing modalities, which may introduce noise, especially when the number of complete samples is limited.

Their framework can be summarized as follows:

- Train modality-specific teacher models using all available samples for each modality (so partially observed cases still contribute).
- Train a multimodal student model on the subset of complete cases, using both true labels and the soft labels of the teachers as additional supervision.

They also emphasize that this strategy is not equivalent to simple late fusion: directly combining teacher predictions can perform poorly, whereas KD provides a training signal that improves the student model while still allowing it to learn internal feature-level integration. This perspective is particularly relevant in multi-omics because the teacher stage can exploit larger per-modality sample sizes, while the student stage can learn cross-modal

synergy on the smaller complete subset—reducing the waste of partially observed patients.

Ranjbari et al. [1] adapted the paper of Wang et al. [6] to incomplete multi-omics integration for disease progression prediction (long vs short-term survival) and propose a framework called KD-SVAE-VCDN. Their main design choices include:

- **Supervised VAEs**(SVAEs) as modality encoders, used to learn latent representations for each modality in a setting characterized by high dimensionality and heterogeneous data types.
- **Two-level teachers and a student model**: first-level teachers are trained on individual modalities; second-level teachers are trained on modality pairs; finally, a student model integrates the subset of samples shared across all modalities, using the teachers' softened predictions through distillation losses.
- **Cross-omics correlation modeling via VCDN** (View Correlation Discovery Network): mean and variance vectors from the modality-specific VAEs are used to construct "discovery tensors," which are concatenated and passed to prediction layers to capture cross-modal interactions.

They report that this architecture can leverage limited fully paired samples while still benefiting from incomplete data through the teacher stages, and they evaluate it on disease progression prediction tasks in breast and kidney carcinoma cohorts. It is not clear however if the prediction on unseen samples can be done on impaired datasets.

2.1.4 Pathway-aware architectures and Pathformer

Beyond generic feature-level integration, a growing line of work emphasizes the importance of incorporating biological structure into deep learning models for multi-omics analysis. In particular, pathway-aware architectures organize molecular features according to known biological pathways, thereby improving interpretability, reducing dimensionality, and aligning learned representations with established biological knowledge.

Pathformer [7] is a representative example of this paradigm, combining pathway-level modeling with transformer-based architectures. Instead of treating genes or molecular features as an unstructured input vector, Pathformer aggregates features into predefined biological pathways (e.g., Kyoto Encyclopedia of Genes and Genomes or Reactome pathways). Each pathway is represented as a high-level embedding, shifting the learning prob-

lem from gene space to pathway space. This design reflects the biological assumption that cancer phenotypes arise from coordinated dysregulation of pathways rather than isolated molecular events. Architecturally, Pathformer follows a hierarchical design. At the first level, modality-specific encoders map raw omics features (such as gene expression or DNA methylation values) into pathway embeddings. These encoders are typically constrained by pathway membership masks, ensuring that each pathway representation only depends on its associated genes. This step substantially reduces dimensionality, shrinking from 11,560 genes to 1,497 pathways, while preserving biological coherence. At the second level, a transformer module operates over pathway embeddings, using self-attention to model interactions between pathways and to identify which biological processes are most relevant for the downstream prediction task.

In Pathformer, this interaction modeling relies on a *criss-cross attention* mechanism rather than standard full self-attention. Criss-cross attention decomposes attention into two structured steps: a row-wise attention that models interactions across pathways within each modality, followed by a column-wise attention that captures interactions across modalities for each pathway. By restricting attention to these predefined axes, the model avoids full quadratic attention while explicitly encouraging biologically meaningful cross-pathway and cross-omics dependencies.

Compared to graph-based integration approaches such as MOGONET [8], Pathformer differs both in representation and in the type of biological structure it exploits. MOGONET constructs patient similarity graphs independently for each omics modality and relies on graph convolutional networks to propagate information across patients, followed by a cross-omics fusion module. While this approach is effective for capturing patient-level relationships and enables biomarker discovery, it fundamentally operates at the sample-similarity level and assumes fully paired multi-omics data.

In contrast, Pathformer focuses on modeling interactions between biological pathways within a patient, rather than similarities between patients, and does not rely on graph construction. As a result, Pathformer offers a complementary inductive bias: pathway-level interpretability and within-sample biological interaction modeling, rather than population-level graph structure.

From the perspective of missing modalities, this distinction is particularly important. MOGONET typically discards samples with missing modalities, which can significantly reduce the effective sample size in real-world multi-omics cohorts. Pathformer, while not

explicitly designed to handle missing modalities, can be more naturally extended to such settings by selectively activating modality-specific encoders or by aggregating pathway embeddings from available modalities only. However, in its standard formulation, Pathformer does not explicitly leverage partially observed samples during training, and therefore does not fully address the unpaired data problem on its own.

These two methods have interesting properties to be compared and to potentially be leveraged in a KD framework.

2.1.5 Aim of this study

While KD-SVAE-VCDN demonstrates that knowledge distillation can effectively leverage incomplete multi-omics data, it does so using latent representations that are not explicitly structured around biological pathways. Conversely, Pathformer provides interpretable, pathway-centered representations but does not natively exploit unpaired samples during training. This complementarity suggests that knowledge distillation could be applied to pathway-aware architectures, allowing biologically structured models to benefit from incomplete data without resorting to explicit imputation.

As a result, one of the objectives of this work is to investigate whether knowledge distillation strategies (conceptually similar to those proposed by Ranjbari et al.) can be adapted to Pathformer's architecture.

Chapter 3

Methodology

For a first approach, we chose to start from the model proposed by Ranjbari et al. [1], since their implementation was available and provided a solid baseline to reproduce. Their framework is designed for incomplete multi-omics settings, explicitly accounting for missing modalities across patients rather than restricting training to fully observed samples.

3.1 Training a Knowledge Distillation model

3.1.1 The data

We used TCGA-BRCA multi-omics data corresponding to the breast cancer cohort composed of mRNA expression (contained in the RNA-seq data), miRNA expression, and DNA methylation, along with clinical survival information. Patients were labeled as short-term vs. long-term survivors, keeping only those with clear labels although a third category "not considered" was available but not considered for now (323 patients after filtering). Because modalities are incomplete for some patients (especially methylation), each modality ends up with a different sample count after alignment with labels.

3.1.2 Our preprocessing strategy compared to Ranjbari et al.[1]

To avoid information leakage, we first performed a stratified 80/20 train–test split on the 323 labeled patients before any feature selection. All statistical selection steps (ANOVA filtering) were fitted using the training set only, and then applied to the test set. This choice differs from the description in Ranjbari et al. [1], where preprocessing and feature selection are described before the split and the leakage-avoidant fitting procedure is not stated explicitly.

Across modalities, we aimed to produce inputs compatible with neural network training and comparable across omics types. Expression matrices (mRNA and miRNA) were log-transformed using $log_2(x+1)$, where x corresponds to the Transcript Per Million (TPM) for mRNA and Reads Per Million (RPM) for miRNA. TPM and RPM normalize raw read counts by gene (or miRNA) length and sequencing depth, and can be written as:

$$\mathrm{RPK}_i = \frac{c_i}{L_i/1000}, \qquad \mathrm{TPM}_i = 10^6 \cdot \frac{\mathrm{RPK}_i}{\sum_j \mathrm{RPK}_j}, \tag{3.1}$$

for mRNA, where c_i is the read count mapped to gene i and L_i is the gene length in base pairs, and as:

$$\mathrm{RPM}_i = 10^6 \cdot \frac{c_i}{\sum_j c_j}, \tag{3.2}$$

for miRNA, where c_i is the read count mapped to miRNA i and $\sum_j c_j$ is the total number of mapped reads in the sample. The matrices were then rescaled to $[0, 1]$ with Min-Max scaling. For methylation, we preserved the same $[0, 1]$ convention (β values are naturally bounded) so that all modalities follow a consistent range. These steps follow the general scaling choices described by Ranjbari et al. [1], but we provide the exact transformations applied in our pipeline.

3.1.2.1 miRNA preprocessing

miRNA data exhibited strong sparsity (many zero or missing values), so we first removed miRNAs with zero expression across all samples, reducing the feature space from 1,881 to 1,598. After applying the $log2(x+1)$ transformation and Min–Max scaling, we aligned miRNA samples with survival labels, resulting in 316 labeled patients with miRNA available (252 in training, 64 in test). Then, the feature selection was performed on the training set using an ANOVA F-test and a p-value threshold of 0.05, which produced 111 retained miRNAs. This is consistent with the ANOVA-based filtering strategy described by Ranjbari et al. [1], although their manuscript reports a different final number of miRNA features and does not specify the split-before-selection constraint. It has to be noted that the input datasets are not necessarily the same as our version of the dataset could differ from theirs.

3.1.2.2 RNA-seq preprocessing

For RNA-seq, we removed genes with zero expression across all samples (60,660 to 57,990) after reshaping the data into a patient-by-gene matrix. We then applied the $log2(x + 1)$ transformation and Min–Max scaling, followed by a variance filter to remove low-variance genes using a 0.02 threshold, retaining 14,915 genes. After aligning with labels, 322 labeled patients had RNA-seq available (257 train, 65 test). Finally, we applied ANOVA-based feature selection on the training set ($p \leq 0.05$), resulting in 1,854 retained genes. These steps closely match the general preprocessing described by Ranjbari et al.[1] (variance filtering at 0.02 and ANOVA filtering), but our reported retained dimensions differ, likely due to implementation details and the explicit leakage-controlled ordering and also the difference in their version of the dataset compared to ours, as mentioned previously.

3.1.2.3 DNA methylation preprocessing

DNA methylation presented the greatest challenge due to extreme dimensionality (486k CpG sites) and missingness. We first filtered CpGs by completeness, keeping only sites with at least 80% observed values across patients (486,427 to 403,114). Remaining missing values were imputed using KNN imputation (K=5, distance-weighted), removing NaNs entirely. After applying the same scaling convention, we aligned methylation with labels and obtained 230 labeled patients with methylation available (186 train, 44 test), making methylation the most incomplete modality in our dataset. For feature selection, we used Welch's ANOVA (robust to unequal variances) on the training set and applied Benjamini–Hochberg multiple testing correction, retaining CpGs with FDR < 0.10, which resulted in 9 selected CpG sites. This part diverges the most from Ranjbari et al.[1]: while they mention scaling, variance filtering, and ANOVA-based filtering for methylation, they do not specify how missing values are handled and report a substantially larger methylation feature set, whereas our pipeline explicitly addresses missingness and controls for multiple testing given the very large number of CpG hypotheses.

3.1.2.4 Preprocessing outcome

After filtering to labeled patients, the dataset contained 323 patients with an imbalanced label distribution (251 long-term vs 72 short-term), and the stratified split preserved these proportions (258 train and 65 test). Following preprocessing and selection, the final dimen-

sionality was reduced to 111 miRNAs, 1,854 mRNA genes, and 9 CpG sites, corresponding to a reduction from 548,968 raw features to 1,974 selected features overall. Importantly, the number of labeled patients differs by modality (316 for miRNA, 322 for RNA-seq, and 230 for methylation), which creates a realistic incomplete multi-omics setting and motivates our later use of distillation frameworks that can exploit all available single-modality samples while remaining robust when modalities are missing.

In the original KD-SVAE-VCDN implementation by Ranjbari et al. [1], the preprocessing strategy appears deliberately stringent, likely driven by practical computational constraints, as the experiments were conducted on limited hardware resources (a single Intel i5 CPU with 8 GB RAM). Hence, our methylation preprocessing pipeline will need to be redesigned, as retaining only 9 CpG sites is likely too restrictive for our downstream objectives; we will therefore explore alternative filtering and selection strategies to increase the methylation feature set, and we discuss these future approaches in the Discussion 5.1 section.

Table 3.1: Summary of Preprocessed BRCA Multi-Omics Dataset

Modality	**Original Features**	**Final Features**	**Train Samples**	**Test Samples**	**Feature Selection Method**	**Normalization**
miRNA	1,881	111	252	64	ANOVA ($p \leq 0.05$)	Log_2 + MinMax
RNAseq	60,660	1,854	257	65	Variance + ANOVA ($p \leq 0.05$)	Log_2 + MinMax
DNA Methylation	486,427	9	186	44	Welch's ANOVA + FDR ($p \leq 0.10$)	KNN imputation + MinMax
Combined	**548,968**	**1,974**	**257**	**65**	–	–

Class distribution: Short-term survival 22.6% (58 train / 14 test), Long-term survival 77.4% (199 train / 51 test).

3.1.3 The KD-SVAE-VCDN model used by Ranjbari et al. [1]

To explicitly handle missing modalities, we implemented the multi-stage KD-SVAE-VCDN framework proposed by Ranjbari et al.[1] (Figure 3.1). As detailed in Section 2 (Litterature Review), the approach relies on a teacher–student distillation strategy built on SVAE encoders and a final VCDN fusion module; here we focus on how the different components are trained and connected in our pipeline.

In practice, the procedure is organized into three stages. First, three level-1 teachers are trained independently, one per modality (mRNA, miRNA, methylation), using all samples available for that modality; each teacher outputs logits that are later used as soft targets for distillation. Second, three level-2 teachers are trained on paired modalities (mRNA–miRNA, mRNA–methylation, miRNA–methylation) using the intersection of

patients available for each pair, while distilling knowledge from the corresponding level-1 teachers. Finally, the student model is trained on the complete subset of patients (all three modalities available); it integrates the modality-specific latent representations through the VCDN module and is optimized with a combination of supervised prediction loss and distillation losses from the level-2 teachers (weighted by coefficients a, b, and c).

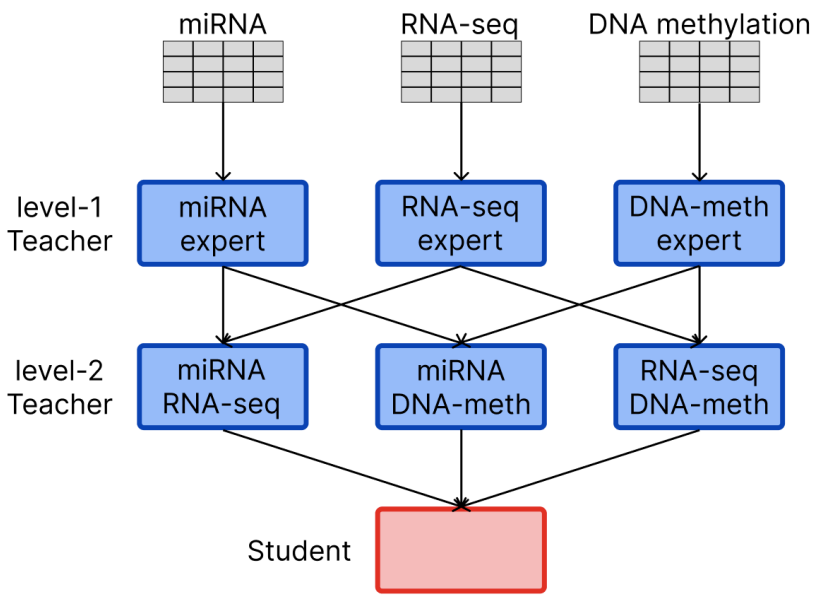


Figure 3.1: Knowledge distillation framework for incomplete multi-omics integration (KD-SVAE-VCDN)

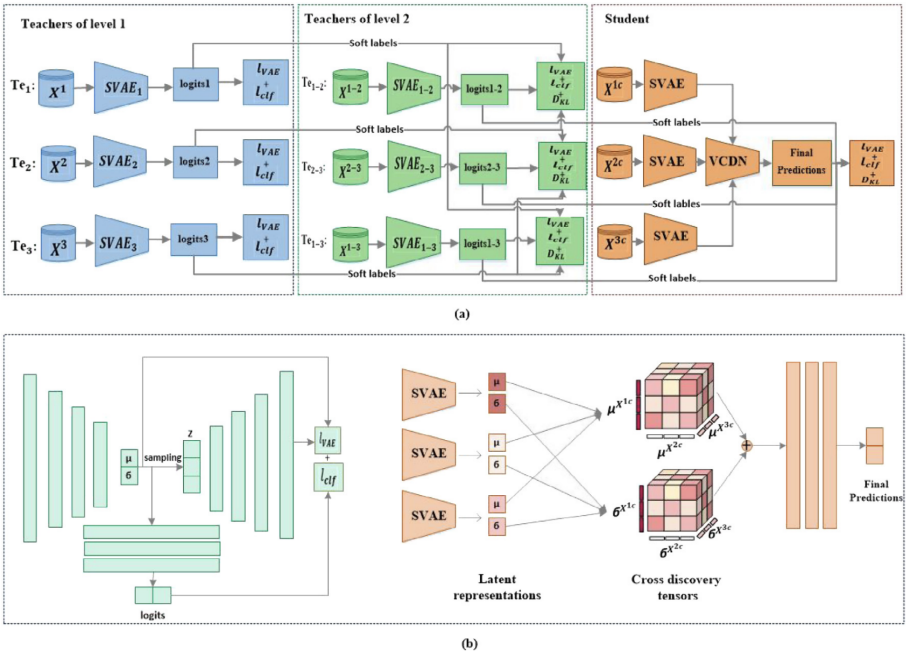


Figure 3.2: Detailed KD-SVAE-VCDN architecture proposed by Ranjbari & Arslanturk. (a) shows the teacher models in each level along with the student model, (b) represents the architecture of the SVAE model (left) and the architecture of the student model (right).

3.2 Pathformer retraining

3.2.1 Original software

The original Pathformer model was implemented using Python 3.6.9 and the PyTorch deep learning framework. PyTorch was used to construct all neural network components, including the pathway-based sparse neural network (PSNN), the Transformer encoder with criss-cross attention, and the fully connected classification layers. Model training, validation, and inference were all conducted within this PyTorch environment.

Model development and experimentation followed a standard scientific Python ecosystem. Data preprocessing and feature construction for multi-omics inputs were performed using custom Python scripts built on commonly used numerical and data-handling libraries (e.g. NumPy and pandas).

Training and evaluation were carried out using 5-fold cross-validation, repeated twice with shuffled splits, fully implemented within Python. Hyperparameter optimization—including learning rate, dropout probabilities, attention coefficients, and training epochs—was performed exclusively on the training folds to avoid information leakage. Performance evaluation metrics (AUC, macro-averaged F1, weighted F1) were computed using Python-based

evaluation routines integrated into the training pipeline.

The complete source code for data preprocessing, model training, evaluation, and benchmarking against comparison methods provided by the authors Wang et al.[8] was released as an open-source package and made publicly available via GitHub, ensuring reproducibility of all experiments reported in the original study.

We were able to run the model training on a copy of the original data, although we did not optimize hyperparameters or run cross-validation for now, since the purpose is to prepare for our new dataset.

We also switched to more recent software, using python 3.11 and pytorch 2.9.1.

3.2.2 Original data from the paper

The original Pathformer study evaluated the model using both tissue biopsy and liquid biopsy multi-omics datasets collected from publicly available sources.

For tissue-based experiments, the authors used multi-omics cancer datasets from TCGA. These datasets covered multiple cancer types and were used for three primary supervised learning tasks:

1. cancer survival risk classification (low- vs. high-risk)
2. cancer stage classification (early stage I–II vs. late stage III–IV)
3. clinical drug response prediction.

Across these tasks, the study included 10 TCGA datasets for survival prediction, 10 datasets for stage classification, and 5 datasets for drug response prediction. The pan-cancer analyses integrated samples from up to 33 TCGA cancer types, depending on task availability.

Each TCGA tissue dataset consisted of matched multi-omics measurements, including:

- RNA expression (gene-level)
- DNA methylation (fragment-level CpG features)
- DNA copy number variation (CNV) (fragment- and gene-level features).

These omics layers were processed into multiple biological modalities using standardized bioinformatics pipelines, producing structured multi-modal inputs aligned at the gene level prior to model training.

All datasets used in the original study were obtained from public repositories (TCGA and Gene Expression Omnibus) and were used exclusively for academic research purposes. No private or proprietary data were included.

3.2.3 Adaptation to new version of the dataset

Although adapting Pathformer to a different version of the dataset and without the CNV omic dataset could in principle involve re-curation or retraining of pathway definitions based on updated genome annotations, we instead reuse the original pathway structures and align genes from the new dataset to these pathways using Ensembl gene identifiers.

Genes in the new dataset were mapped to the original gene–pathway associations via Ensembl ID matching.

We rely on the same pathway definitions and gene–pathway mapping tables as in the original Pathformer implementation. These biological resources are known to evolve slowly over time, as they encode stable biological knowledge, with changes mainly reflecting updates in annotation versions rather than fundamental modifications of pathway structure. Using consistent pathway and gene–pathway tables therefore ensures biological coherence and facilitates comparability with previously reported results.

While our study leverages updated omics datasets and excludes CNV data, this does not require changes to the underlying Pathformer framework. The model's design allows for flexible integration of heterogeneous omics inputs, enabling it to accommodate variations in data availability without altering the pathway topology. As a result, the framework can exploit updated gene-level measurements while preserving interpretability and avoiding confounding effects related to changing pathway definitions.

3.3 Hardware

So far, we performed the training of Pathformer on a single NVIDIA RTX 2080ti 11GB GPU. The original Pathformer paper used a single NVIDIA A800 80 GB GPU.

We plan to switch to the DCE cluster [9] for the hyperparameter tuning and the distillation steps.

Chapter 4

First Results

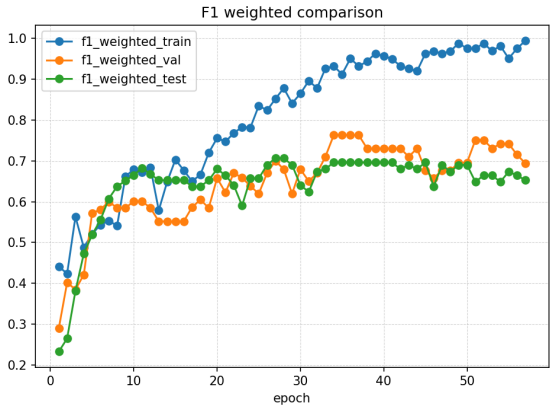
4.1 Implementation of the KD-SVAE-VCDN model

At this stage, we have implemented the full preprocessing pipeline for the three modalities and begun adapting the KD-SVAE-VCDN codebase proposed by Ranjbari et al.[1] to our TCGA-BRCA setting. However, no reliable or interpretable modeling results have been obtained yet, as the training and evaluation are still ongoing. In parallel, we are actively redesigning the DNA methylation preprocessing and feature selection strategy, since the current pipeline retains too few CpG sites to support meaningful downstream learning.

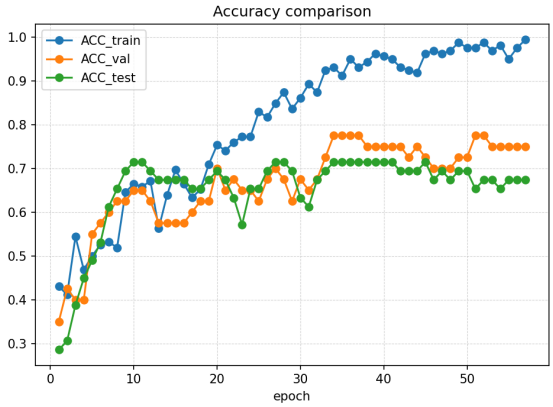
4.2 Pathformer

While adapting to the new version of the dataset, we performed a re-training of Pathformer on its original data, to test our hardware capacities and software compatibilities.

First results for the accuracy and F1-weighted criteria have been obtained and are displayed in 4.1. This very basic and raw re training clearly overfits pretty fast. We will take into account the steps used for Pathformer's SOTA performance when implementing it on our dataset (hyperparameter optimization).



(a) F1 weighted



(b) Accuracy

Figure 4.1: Model performance first comparison

Chapter 5

Discussion and Perspectives

5.1 Discussion and perspectives

Summary of the current progress. At this stage, we have implemented a complete preprocessing pipeline for TCGA-BRCA multi-omics data, started the first implementations/adaptations of the KD-SVAE-VCDN framework proposed by Ranjbari et al.[1], and in parallel reproduced the Pathformer architecture from its original codebase as a basis for future experiments and distillation perspectives.

KD-SVAE-VCDN: main observations and limitations. A main outcome of this first implementation phase is that **DNA methylation preprocessing remains the bottleneck**. While the expression modalities (RNA-seq and miRNA) yield reasonably sized and stable feature sets after filtering and ANOVA-based selection, the methylation pipeline as currently implemented results in an extremely small set of retained CpG sites (9), which is unlikely to capture enough biological signal for downstream representation learning and multimodal fusion. This indicates that methylation requires a dedicated redesign to better manage its very high dimensionality while retaining an informative yet computationally tractable feature space.

A second point concerns **reproducibility and methodological precision**. Although Ranjbari et al.[1] provide a strong conceptual framework for learning under incomplete modalities, several preprocessing steps are described at a high level, with limited operational detail (ordering of split vs. selection, handling of missing methylation values, and how thresholds lead to the reported final feature counts). This ambiguity makes strict reproduction non-trivial and partly motivated our more explicit, leakage-aware pipeline (train/test split before feature selection and selection fitted on training only). Overall, im-

proving transparency in preprocessing choices is essential because these steps materially affect learned representations and the performance attributed to the distillation strategy itself.

Perspectives for KD-SVAE-VCDN. In the next phase, we will iterate on the methylation preprocessing to obtain a more informative CpG feature set (revisiting completeness thresholds, selection criteria, and alternatives to overly aggressive multiple-testing filtering), and re-train KD-SVAE-VCDN with these revised inputs to evaluate training stability and predictive performance under a more robust methylation representation. Beyond reproducing the baseline, we also plan to evaluate an alternative architecture (Figure 5.1) to test whether different integration mechanisms can better exploit the distilled knowledge and mitigate remaining modality imbalance.

Pathformer: current status and perspectives. In parallel to the KD-SVAE-VCDN track, the Pathformer architecture has been reproduced using the original codebase and experimental setup, which ensured a correct understanding of the pathway-aware transformer design and established a reliable reference implementation. No retraining on a new cohort has been performed so far.

The next phase will focus on adapting the selected TCGA multi-omics cohort to Pathformer requirements (notably survival label definition and potential genome annotation version discrepancies), followed by retraining Pathformer to obtain a fully trained pathway-aware model. This model will serve both as a strong biologically informed baseline and as a natural candidate teacher in subsequent knowledge distillation experiments.

Perspectives: distilling a Pathformer teacher. Inspired by the framework of Ranjbari & Arslanturk [1], we will then investigate distillation strategies where a trained Pathformer teacher is distilled into student models operating on reduced modality subsets. The objective is to produce a modality-adaptive predictor robust to missing views at inference time, while preserving as much pathway-level knowledge learned by the teacher as possible. Finally, we will compare this pathway-aware distillation approach against (i) the Pathformer teacher, (ii) KD-SVAE-VCDN trained with the improved preprocessing, and (iii) non-distilled baselines, in order to quantify the trade-off between biological inductive bias (interpretability) and robustness to incomplete multi-omics inputs.

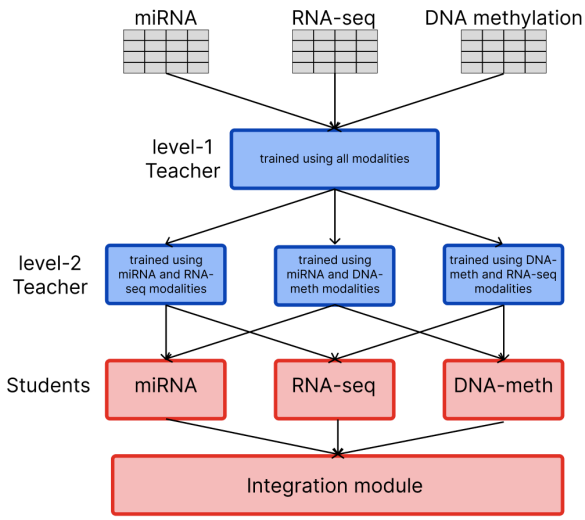


Figure 5.1: Proposed alternative multi-omics architecture for survival prediction under missing modalities.

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